Shri Guru Gobind Singhji Institute of Engineering and Technology
Vishnupuri, Nanded (Maharashtra State) INDIA PIN 431606
Government Aided Autonomous Institute DTE Code: 2020
NAAC Accredited institute GRADE B++, CGPA 2.91 (2020 -2025)
Vision Statement: Education of Human Power for Technological Excellence

"FIRE ALARM CIRCUIT"

For the Second Year Micro Project

In

Electrical Engineering Department

Project Report

By

PGEE21

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Under the Guidance of

Prof. S.S. Jamraj

Date: 22/04/2025

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CERTIFICATION

This is to certify that the project entitled "Fire Alarm Circuit" in the fulfilment of "Second Year Micro Project" for Shri Guru Gobind Singhji Institute of Engineering and Technology, Vishnupuri, Nanded. This Bonafide work carried and completed under guidance and supervision of our Prof. Satish Jamraj during academic year 2024-2025.

SUBMITTED BY: PGEE21

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Project Guide

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Declaration by Students

We hereby declare that the project titled **"Fire Alarm Circuit"** has been carried out by us as a part of our academic curriculum and is the result of original work undertaken by us. The circuit has been designed, assembled, and tested under the expert guidance of **Prof. S.S. Jamraj**, to whom we are sincerely grateful for his constant support and valuable insights. We affirm that this project does not contain any material copied from other sources without due acknowledgment. We also take this opportunity to thank **Dr. S.M. Gudhe**, Head of Department, for providing us with the necessary facilities, encouragement, and a conducive environment for completing this project successfully.

Mr. Mrunal Bhamare	2023BEL026	……..
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We would like to express our sincere gratitude to all those who have contributed to the successful completion of this fire alarm circuit project. First and foremost, we extend our heartfelt thanks to **Prof. S.S. Jamraj**, our project guide, for his invaluable guidance, encouragement, and constant support throughout the course of this work. His expertise and insights were crucial in helping us overcome various challenges and complete the project efficiently.

We are also immensely grateful to **Dr. S. M. Gudhe**, Head of the Electrical Engineering Department, for providing us with the necessary resources and a conducive environment for learning and experimentation. His leadership and continuous motivation played a significant role in shaping the quality and direction of this project.

A special mention must be made of all the faculty members of the **Electrical Department**, whose timely feedback, technical input, and mentorship helped enhance our understanding of core concepts and practical implementation. Their dedication to teaching and mentoring students has been truly inspiring.

Lastly, we would like to acknowledge the support of the **AICTE IDEA Lab**, which provided access to essential tools, components, and infrastructure required for the successful development and testing of our fire alarm circuit. The hands-on experience and exposure we received at the lab played a pivotal role in refining our technical and problem-solving skills.

ABSTRACT

This project focuses on the design and implementation of a basic fire alarm system intended for early detection of fire hazards through heat sensing. The primary objective is to create a low-cost, easy-to-build circuit that can alert individuals in case of abnormal temperature rise, helping prevent property damage and ensuring safety in homes, classrooms, and small offices.

The working principle of the system is based on the behaviour of a thermistor, which changes its resistance in response to temperature. As the surrounding temperature increases, the thermistor's resistance decreases, leading to a change in voltage across the circuit. This voltage variation is then processed through a series of transistors that act as amplifiers and switches. A timer integrated circuit is used to generate an output signal once the heat threshold is crossed. Supporting components such as resistors, capacitors, and diodes help to condition the signal, stabilize the operation, and protect the circuit from voltage fluctuations.

When the thermistor is exposed to a heat source (such as a lighter during testing), the system is triggered, causing both a buzzer and an LED to activate. This provides immediate audio-visual feedback indicating a possible fire hazard. The entire setup is powered by a small DC voltage source and is designed for portability and simplicity.

The circuit performed successfully during testing, with prompt activation upon detecting heat. Its low power requirement, minimal component count, and effectiveness make it a practical and educational solution for basic fire detection applications. This project not only demonstrates fundamental electronics knowledge but also provides real-world utility in enhancing fire safety.

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CHAPTER 1
INTRODUCTION

1.1 BACKGROUND

Fire hazards pose a significant threat to life and property, making early detection systems essential in ensuring safety and timely response. Traditional fire alarms are often expensive and complex, limiting their accessibility in small-scale or residential settings. This project aims to design a simple, cost-effective, and efficient **fire alarm circuit** using basic electronic components such as a **thermistor, 555 Timer IC, transistors, resistors, capacitors, diode, LED, and buzzer**. The thermistor, a temperature-sensitive resistor, plays a crucial role in detecting abnormal temperature rises caused by fire. The circuit activates an alarm system when a threshold temperature is reached, alerting individuals to take prompt action. By using a 555 timer in astable mode and simple transistor switching, this design ensures both reliability and affordability. This project not only provides practical knowledge of electronics but also contributes to developing basic safety solutions for everyday environments.

1.2 MOTIVATION

The increasing number of fire-related incidents in homes, offices, and public places highlights the urgent need for early detection systems that are both reliable and affordable. This project was inspired by the desire to create a **simple yet effective fire alarm circuit** that can be easily built, understood, and deployed in everyday environments. We were motivated to explore how basic electronic components like **thermistors, transistors, and the 555 Timer IC** could be used to design a low-cost solution that responds quickly to rising temperatures. Understanding the real-world application of electronics to solve safety-related problems encouraged us to take on this project. It not only strengthened our technical knowledge but also gave us the opportunity to contribute to a socially relevant issue — fire safety. Our goal is to promote awareness and the importance of having accessible safety systems, especially in areas where commercial fire alarms are unaffordable or unavailable.

1.3 OBJECTIVES

The primary objective of this project is to design and implement a **low-cost, efficient, and reliable fire alarm circuit** using basic electronic components. The system aims to detect a rise in temperature through a **thermistor** and trigger an immediate **audio & visual alert** using a **buzzer & LED**. Additional goals include understanding the working principles of the **555 Timer IC in astable mode**, transistor switching operations, and the role of passive components in practical circuits. This project also focuses on enhancing hands-on skills in circuit design, simulation, and troubleshooting, while promoting the importance of early fire detection in ensuring safety. The project serves as a stepping stone for developing more advanced fire detection systems in the future.

1.4 SCOPE OF WORK

This project focuses on the development of a **basic fire alarm circuit** capable of detecting elevated temperatures and triggering an **audio alert** through a buzzer. The work involves studying the characteristics of a **thermistor** as a heat sensor, designing a circuit using a **555 Timer IC, transistors, resistors, capacitors, diode**, and integrating them effectively to ensure timely response to heat. The project includes schematic design, simulation, hardware implementation, and functional testing of the circuit. While the current scope is limited to **temperature-based fire detection**, it lays the groundwork for future enhancements such as smoke detection, wireless alerts, or integration with home automation systems. The circuit is intended for use in small spaces like homes, labs, or classrooms where cost-effective safety solutions are essential.

1.5 ORGANIZATION OF REPORT

- **Chapter 1: Introduction**

  Covers the background of the project, motivation, objectives, and the scope of work.

- **Chapter 2: Literature Review**

  Includes a summary of related work, comparisons with existing approaches, and identification of gaps in current systems.

- **Chapter 3: System Design / Methodology**

  Provides the block diagram or flowchart along with a detailed description of the proposed system, system architecture, and hardware/software requirements.

- **Chapter 4: Implementation**

  Discusses the tools used, hardware description, circuit diagrams, PCB layouts (if any), and screenshots or build photos.

- **Chapter 5: Results and Discussion**

  Presents the output of simulation or hardware testing, supported by graphs and tables, followed by comparison with expected results.

- **Chapter 6: Applications and Future Scope**

  Describes real-world applications and suggestions for future improvements or expansions of the system.

- **Chapter 7: Conclusion**

  Summarizes the key achievements of the project and includes final remarks.

CHAPTER 2
LITERATURE REVIEW

2.1 SUMMARY OF RELATED WORK

The evolution of fire alarm systems has been shaped by the growing need for safety in residential, industrial, and commercial spaces. Over the years, numerous types of detection technologies have been explored. Most conventional fire alarm systems use smoke detectors or infrared sensors that rely on the detection of particulate matter or temperature differences in the environment. These systems are often connected to microcontrollers or microprocessors that analyze signals and trigger alarms accordingly. Research studies and industrial applications show that while these systems are efficient, they also tend to be complex and expensive, limiting their accessibility in low-budget or rural areas. In contrast, thermistor-based fire alarm circuits provide a simpler, cost-effective alternative. These systems detect changes in ambient temperature and trigger alarms when a predefined threshold is crossed. The reliability and affordability of such designs have made them a topic of interest in basic safety implementations. Several academic projects and DIY applications have documented successful use of thermistors, transistors, and timer ICs in building fire alarm systems without relying on digital processing.

2.2 COMPARISONS WITH EXISTING APPROACHES

When comparing thermistor-based fire alarm systems to more modern, sensor-driven solutions, several distinctions emerge. High-end fire detection systems often incorporate technologies such as IoT, wireless sensor networks, and artificial intelligence. These systems allow for remote monitoring, early-stage fire prediction, and direct communication with emergency services. While advanced systems are highly efficient and accurate, they come at a significant cost and often require maintenance and technical expertise. On the other hand, thermistor-based systems, like the one developed in our project, offer a straightforward, analog approach to fire detection. They are designed for

small-scale applications where cost, simplicity, and reliability are the primary considerations. The use of passive components such as resistors, capacitors, and basic transistors minimizes complexity while still ensuring timely detection of fire-related temperature changes. However, these systems typically lack advanced features like visual notifications, wireless alerts, or multi-zone detection, which are standard in modern systems. Thus, while both approaches have their merits, the choice depends heavily on application context and resource availability.

2.3 GAPS IDENTIFIED

Despite the advancements in fire alarm technologies, certain practical gaps remain, especially in affordability, scalability, and ease of implementation. Many existing solutions, although feature-rich, are unsuitable for small-scale deployments due to their high cost, dependency on external connectivity, and power consumption. Moreover, integrating and maintaining these systems in low-resource settings such as rural schools, homes, or small shops becomes challenging. The absence of simple, autonomous fire alarm systems that can operate without microcontrollers or internet access is a significant limitation. Our project addresses this gap by designing a fire alarm circuit that operates solely on analog components like a thermistor, 555 timer IC, and transistors. This ensures functionality even in areas with unstable power or no internet infrastructure. However, gaps still persist in the form of limited alert mechanisms — for instance, the absence of a visual indicator in our design may be a drawback in noisy environments. Future iterations could explore modular designs that add visual and wireless alerts while retaining simplicity and low cost.

CHAPTER 3

SYSTEM DESIGN

3.1 BLOCK DIGRAM

The fire alarm circuit is organized into simple yet functional blocks for clarity. The first block involves the power supply which energizes the system. The second block is the sensing unit comprising a thermistor that reacts to temperature changes. Under normal conditions, the thermistor maintains a specific resistance value. When the temperature rises beyond a predefined threshold, the thermistor's resistance drops, altering the voltage across a transistor, which activates the trigger circuit. The signal processing block consists of a 555 timer IC configured in astable mode to produce a pulse that drives the output. This output is connected to a buzzer & LED, forming the alarm block. A LED indicates power status, enhancing circuit monitoring. The flowchart begins with powering the system, monitoring temperature, detecting threshold breach, and finally triggering the buzzer. This logical sequence ensures easy understanding of the system's operation for anyone replicating or expanding the project.

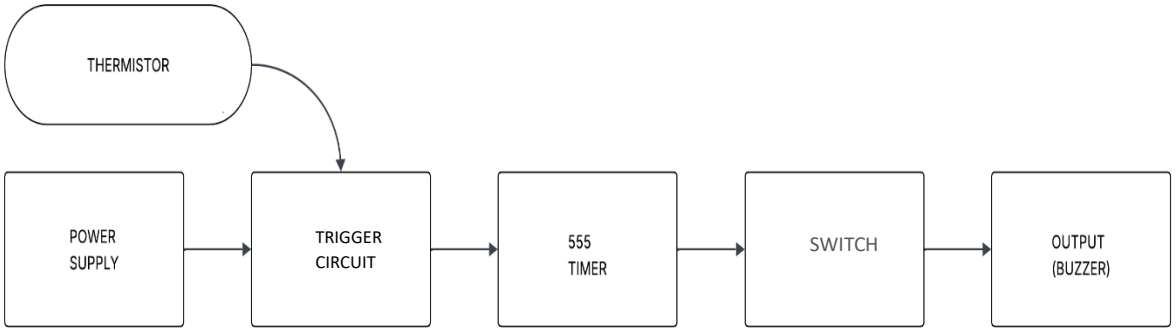


Fig. 3.1 Block Diagram

3.2 DESCRIPTION OF PROPOSED SYSTEM

The proposed fire alarm system uses a thermistor to sense heat. It is designed to detect a rapid increase in temperature due to fire and activate an audible alert. The circuit uses two BC548 and one BC547 transistor for signal switching and amplification. A 555 timer IC is employed in astable mode to generate the timing pulse that controls the buzzer. Supporting components include resistors to limit current, capacitors for stability and timing, and a diode to prevent reverse polarity. The circuit operates on an 6V DC supply. When the thermistor detects heat, its resistance decreases, which biases the transistors, eventually triggering the 555 timer. This, in turn, activates the buzzer, providing an audible alert. The design is compact and efficient, ideal for low-resource settings. Unlike modern fire alarm systems, this project also includes a visual alert (like flashing LEDs), along with this it also focuses on sound-based notification for immediate human response.

3.3 SYSTEM ARCHITECTURE

The system architecture of our fire alarm circuit consists of interconnected hardware components designed to perform temperature-based fire detection. The input stage is the thermistor, which varies its resistance with temperature. This resistance change is converted to a voltage signal, processed by a series of transistors acting as switches and amplifiers. The core processing component is a 555 timer IC configured in astable mode, which is responsible for generating a fixed-duration output pulse upon receiving a trigger. This output drives the speaker or buzzer through another transistor. Power regulation is ensured using resistors and capacitors, while signal conditioning is aided by a diode to prevent reverse feedback. The output stage is a buzzer that produces an audible alarm. The circuit's simplicity allows it to work without programming or digital interfacing. This architecture ensures fast response time, reliable operation, and minimal component failure.

3.4 HARDWARE / SOFTWARE REQUIREMENTS

Hardware



Fig 3.2 Thermistor



Fig 3.3 555 Timer



Fig 3.4 Transistors

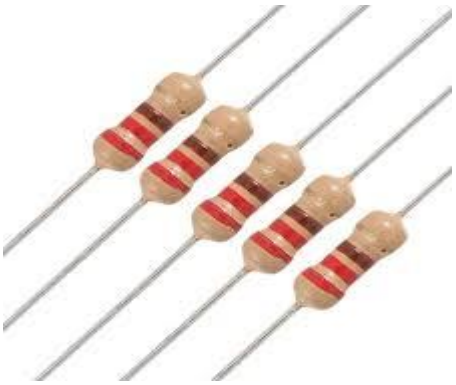


Fig 3.5 Resistors



Fig 3.6 Capacitors



Fig 3.7 Diode



Fig 3.8 Battery



Fig 3.9 LED



Fig 3.10 Buzzer

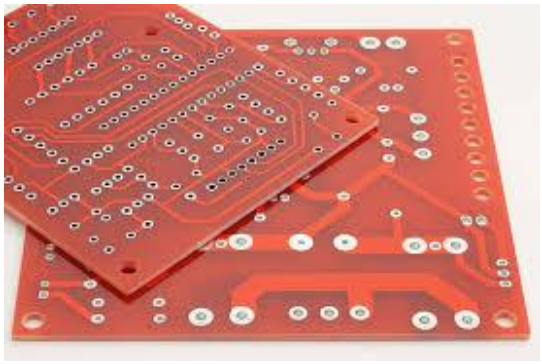


Fig 3.11 PCB

Software

- Proteus (for circuit simulation)
- KiCad (for PCB design)

CHAPTER 4

IMPLEMENTATION

4.1 TOOLS USED

To implement our fire alarm circuit, a mix of simulation and physical hardware tools were employed to ensure both theoretical validation and practical functionality. For circuit simulation and testing, **Proteus Design Suite** was used. It allowed us to design the entire circuit virtually, test its real-time behaviour, and make corrections without wasting physical components. For PCB design, **KiCad** was chosen due to its open-source nature and comprehensive feature set for creating single-layer PCB layouts. Once the virtual model was validated, physical tools such as a **breadboard**, **multimeter**, **soldering iron**, and **power supply unit** were used. Basic tools like **wire cutters**, **strippers**, and **probes** were also necessary during assembly. These tools helped in checking connections, measuring voltages across components, and ensuring current flow in the circuit was as expected. The seamless integration of digital tools with traditional electronic equipment helped streamline development, reduced trial-and-error cycles, and made the overall build process more efficient and accurate.

4.2 HARDWARE DESCRIPTION

This project is a pure hardware implementation and does not involve any software programming. The design revolves around an analog configuration that combines a **thermistor**, **transistors**, and a **555 timer IC** to detect high temperatures and activate an alarm. The thermistor, a temperature-sensitive resistor, is responsible for detecting sudden heat increases. Under normal room temperature, its resistance remains high, keeping the circuit inactive. However, when a fire-induced temperature rise occurs, its resistance drops, allowing a voltage shift that biases the base of a **transistor**. This activates the next stage of the circuit, triggering the **555 timer** configured in astable mode. Once triggered, the timer generates a high output for a brief duration, powering a **buzzer** that serves as the audible alarm. A **diode** is used to prevent reverse polarity, and **resistors** and **capacitors** help set the sensitivity and timing of the system. **LEDs** provide a basic power indicator. All components

were finally assembled on a **single-layer PCB**, making the build durable and compact

4.3 CIRCUIT DIAGRAM / PCB LAYOUTS

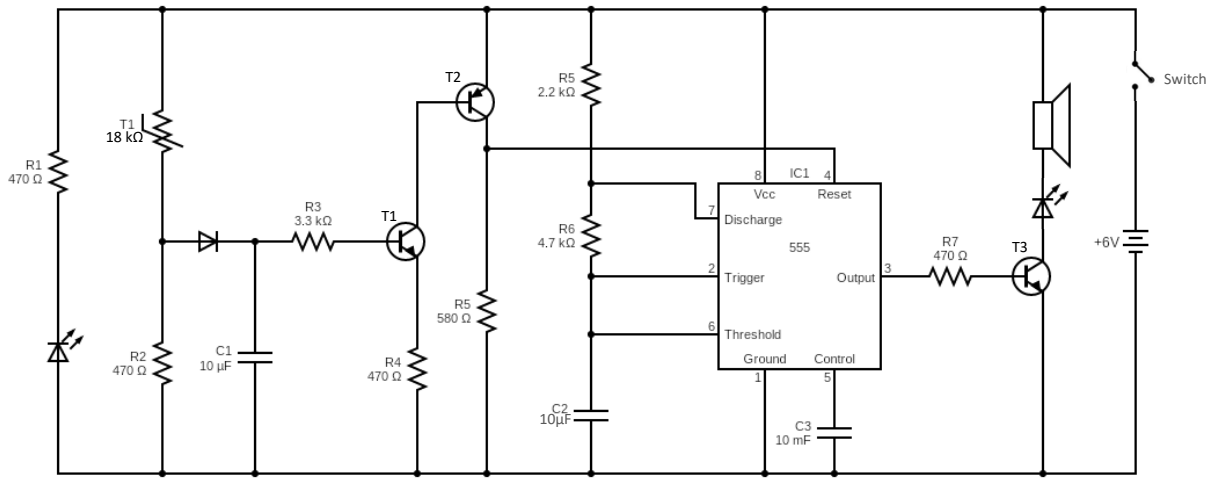


Fig 4.1 Circuit Diagram

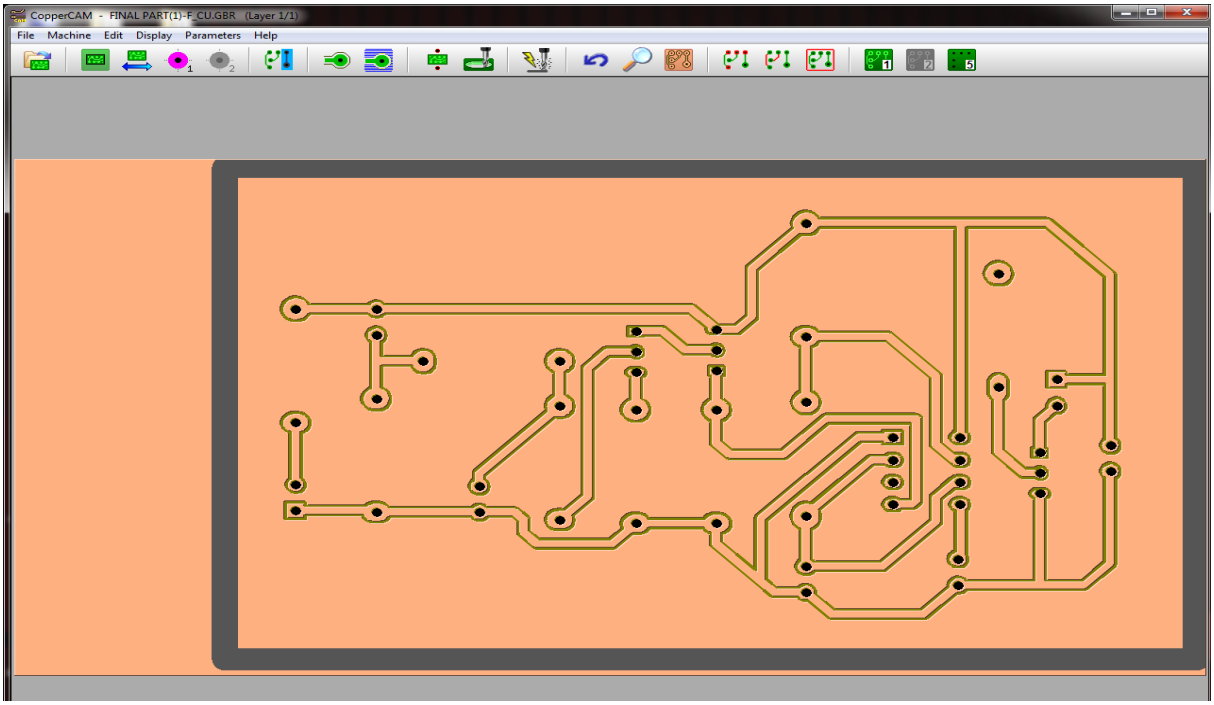


Fig 4.2 PCB Layout

4.4 SCREENSHOTS / BUILD PHOTOS

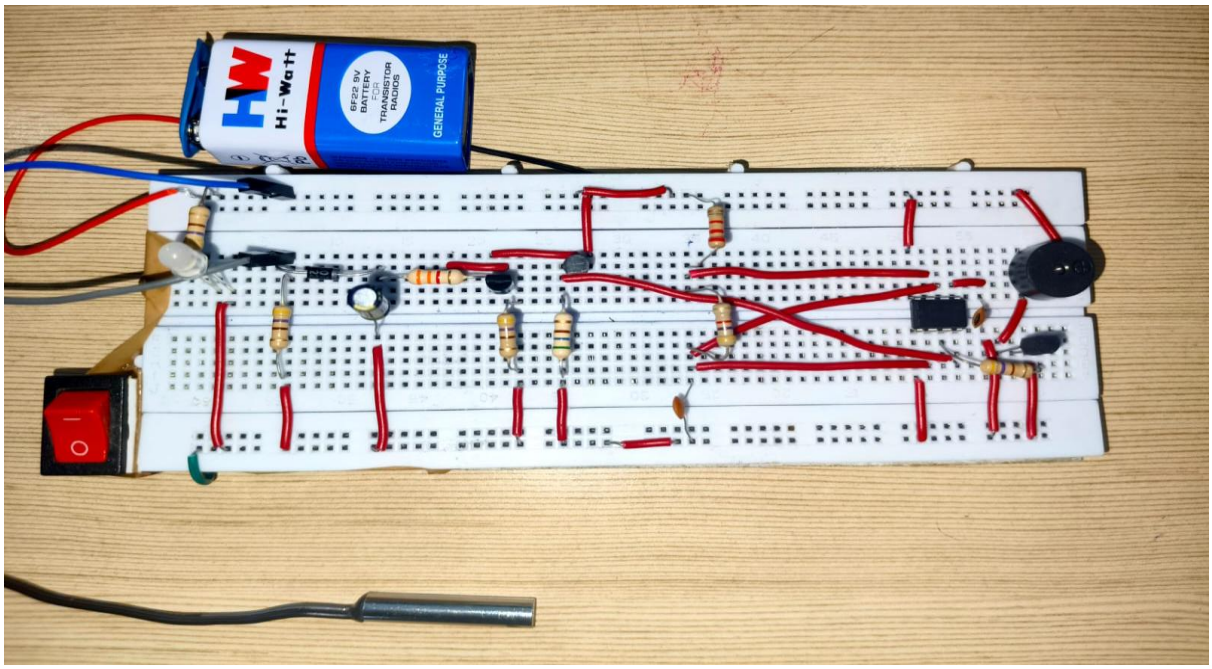


Fig 4.3 Breadboard

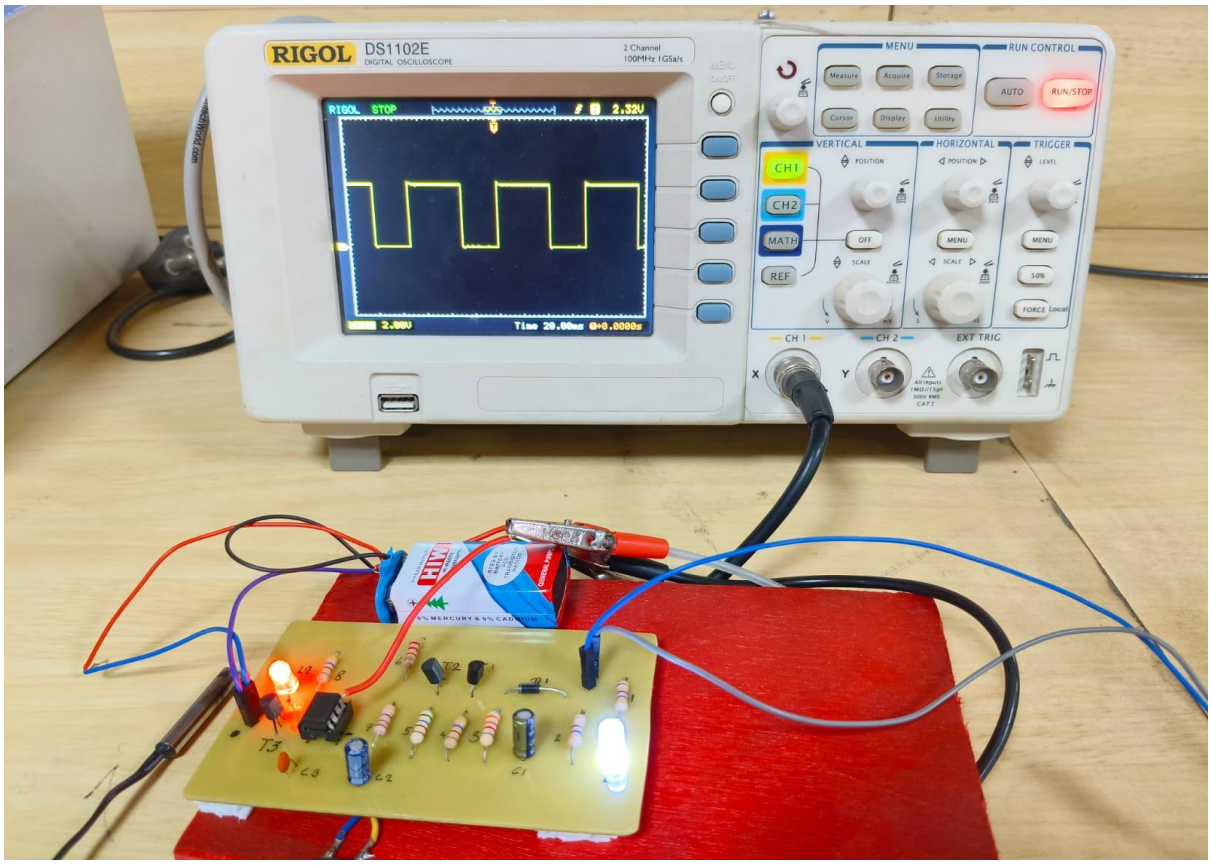


Fig 4.4 PCB Output

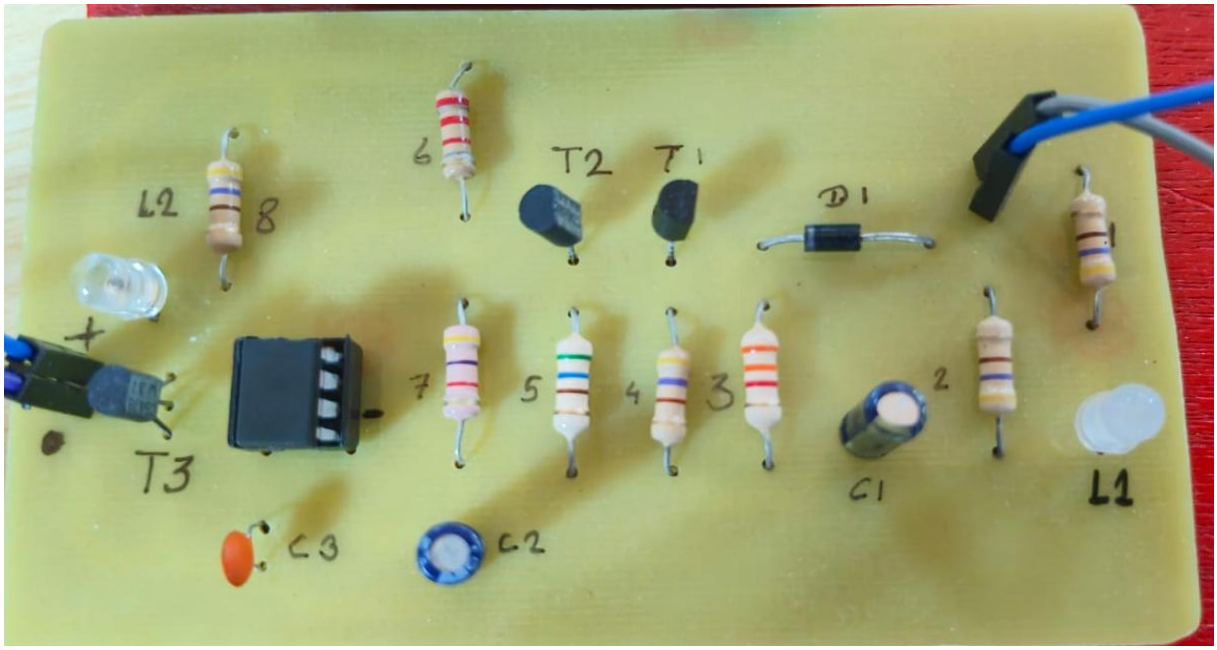


Fig 4.5 PCB (Front)

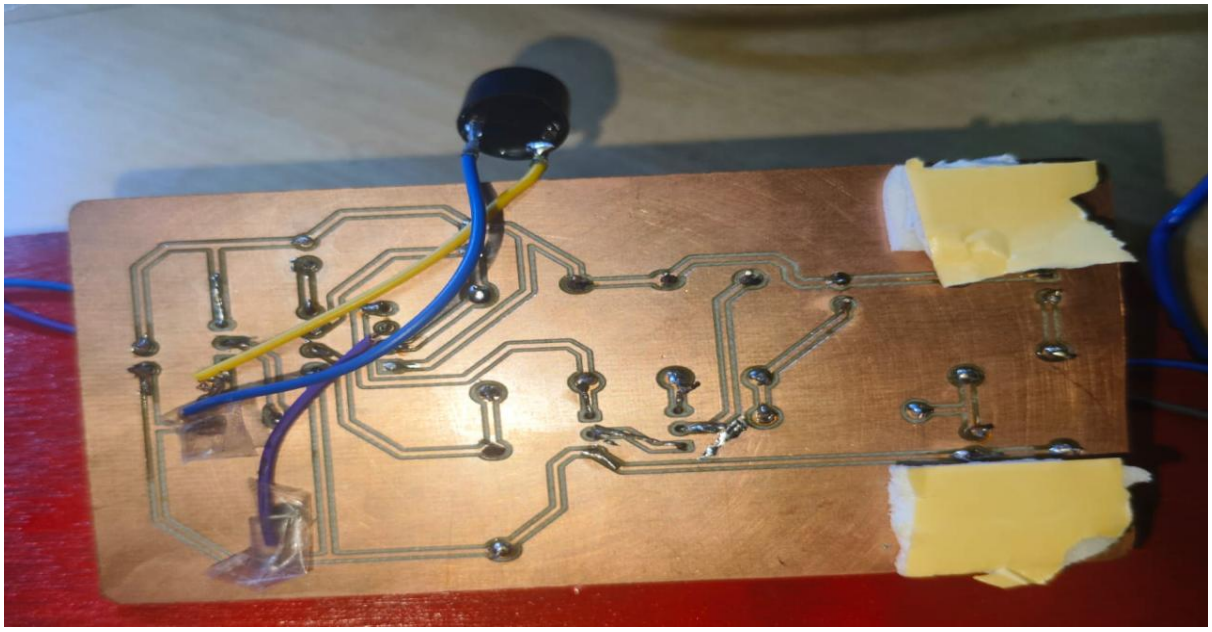


Fig 4.6 PCB (Back)

CHAPTER 5
RESULT & DISCUSSION

5.1 OUTPUT OF SIMULATION / HARDWARE

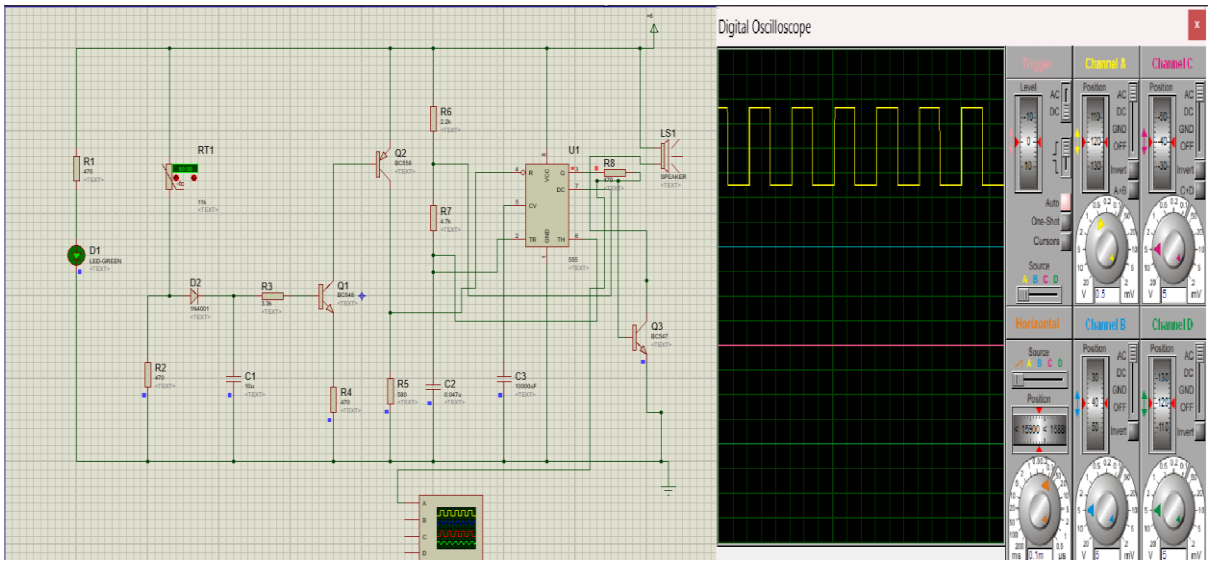


Fig 5.1 Simulation Output

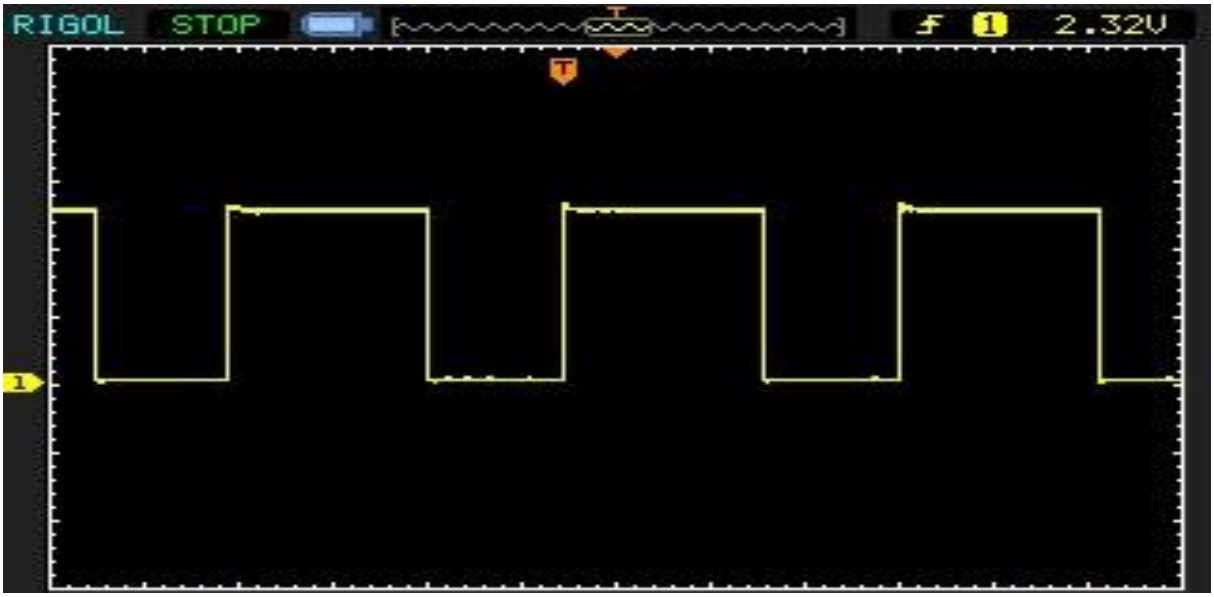
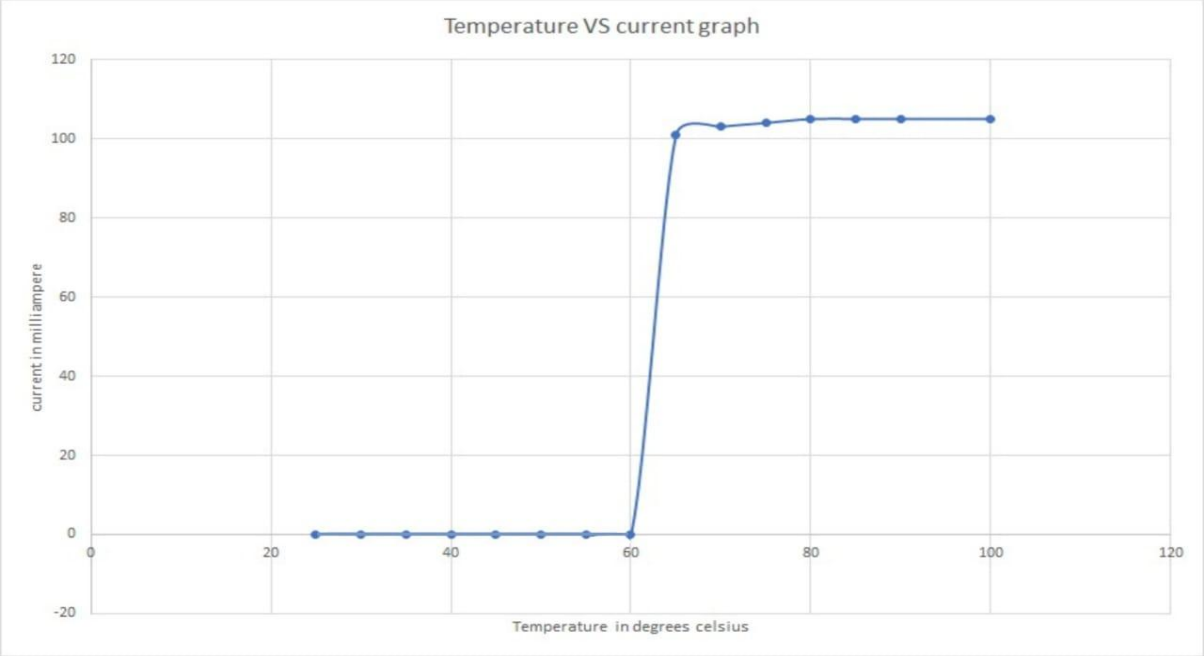


Fig 5.2 PCB Output

5.2 GRAPHS, TABLES AND RESULT INTERPRETATION



X-axis	Y-axis
Temp(°C)	Current(mA)
25	0
30	0
35	0
40	0
45	0
50	0
55	0
60	0

X-axis	Y-axis
Temp(°C)	Current(mA)
65	101
70	103
75	104
80	105
85	105
90	105
100	105

Table 5.1 Readings

Result Interpretation

Charging period of Capacitor (Astable mode) = T1

$T1 = 0.693(Ra+Rb)C$

Ra=2.2kΩ

$Rb=4.7k\Omega$

C=10μF

T 1 =0.693(2.2+4.7)×10³ ×10×10^-6

$T1=0.0483sec$

$T1=47ms$

Discharging period of Capacitor = T2

$T2=0.693RbC$

$T2=0.693\times4.7\times10^{3}\times10\times10^{\wedge}-6$

$T2=0.0325sec$

T2= 32 ms

Duty Cycle %=T(on)/T(total)×100

D.C = 0.693(Ra+Rb)C/0.693(Ra+2Rb)C ×100

D.C= Ra+Rb/Ra+2Rb×100

=2.2+4.7/2.2+2×4.7×100

=6.9/11.6×100=59.48%

D.C=59.48%

5.3 COMPARISON WITH EXPECTED RESULTS

The fire alarm circuit was designed using a 555 Timer IC operating in astable mode, aimed at generating a continuous square wave signal. This output serves to activate the alert components—specifically a buzzer and an LED—during a fire trigger scenario. The expected behaviour was the generation of a steady square wave, which would in turn drive the LED to blink and the buzzer to beep rhythmically. After implementing the circuit on a PCB, the output waveform was observed using an oscilloscope. The result, as seen in the captured image, shows a clean and consistent square wave with minimal distortion, aligning well with theoretical expectations. The LED and buzzer both responded accordingly: the LED blinked at a steady rate while the buzzer emitted periodic beeps, indicating proper signal delivery from the 555 Timer. The use of a PCB contributed to stable operation, reducing issues such as loose connections or noise often seen in breadboard setups. Overall, the observed results validate the design's accuracy and functional integrity.

EXPECTED

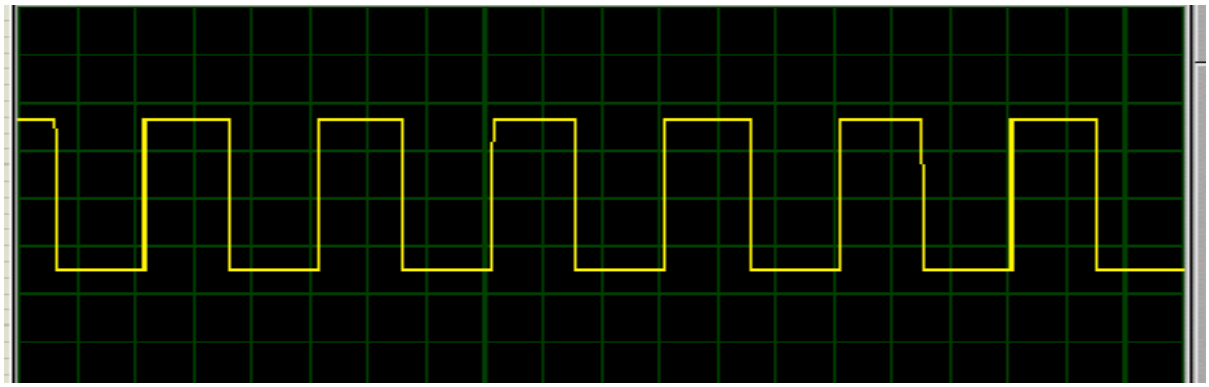


Fig 5.3 Expected Output

OUTPUT

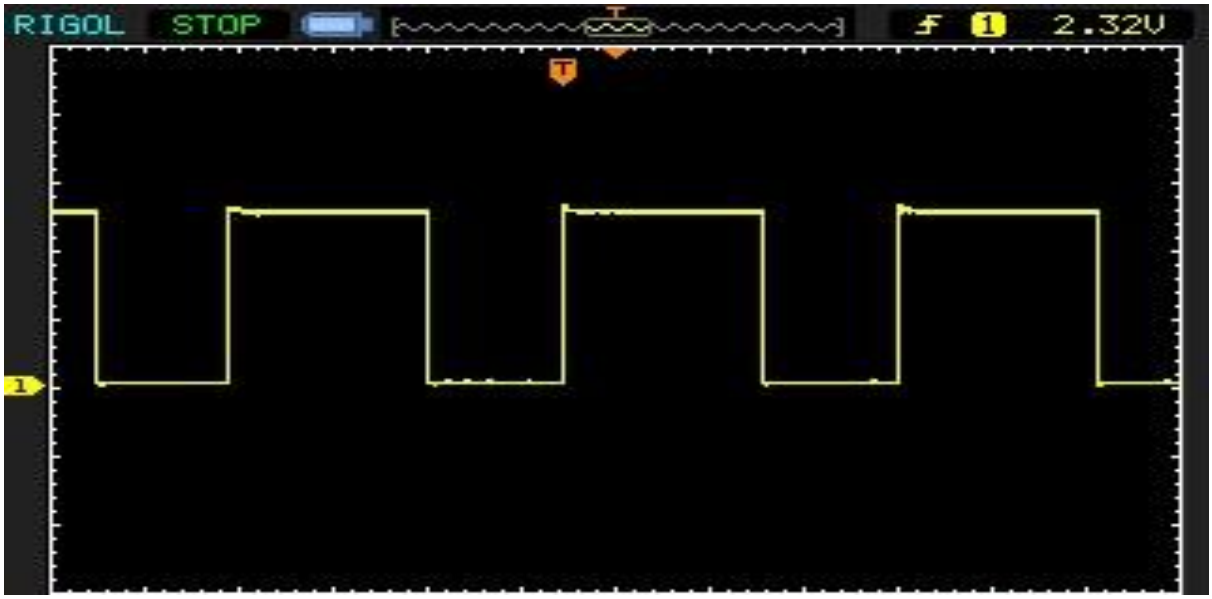


Fig 5.4 PCB Output

CHAPTER 6
APPLICATIONS AND FUTURE SCOPE

6.1 REAL WORLD APPLICATION

The Fire Detector System designed using a **555 Timer IC**, **thermistor**, and **LED indicator** is a compact and efficient solution ideal for various practical scenarios. Due to its simplicity, low cost, and reliability, it can be used in a wide range of fire safety applications.

1. Residential Safety

- Used in homes to detect fire hazards in kitchens, living rooms, and electrical areas.
- Provides an early warning through a visual & audio alert, preventing major incidents.

2. Educational Projects

- Suitable for school and college-level electronics or science projects.
- Helps students understand the basics of sensors, transistors, and circuit behaviour.

3. Small Offices and Shops

- Can be installed in small commercial areas where expensive fire alarm systems are not viable.
- Useful in storerooms, server rooms, or areas with electrical equipment.

4. Laboratories

- Detects fire risks in chemistry and electronics labs where experiments or components may overheat.
- Provides a quick alert for preventive action.

5. Warehouses and Storage Units

- Helps monitor temperature-sensitive environments where overheating could lead to fire.

6.2 ENHANCEMENT OR FUTURE SCOPE

1. Wireless Communication Integration

- **Need**: Traditional fire alarms only offer local alerts, limiting their functionality in modern smart homes or industrial IoT applications.

- **Enhancement**: Incorporating a wireless communication module (Wi-Fi, Bluetooth, or GSM) would allow the system to send remote notifications to mobile devices or a cloud service via SMS or app alerts. This integration ensures that the user is informed about the alarm even when not physically present near the device.
- **Benefit**: This makes the system suitable for smart homes, remote monitoring, and industrial applications, ensuring faster response to potential fire hazards.

2. Replacement of Thermistor with a Digital Temperature Sensor

- **Need**: Thermistors may lack the precision and ease of calibration required for highly accurate temperature detection.
- **Enhancement**: Replacing the thermistor with a digital temperature sensor like LM35 or DHT11 would provide more accurate, stable, and programmable temperature measurements. These sensors offer digital outputs, making integration with microcontrollers easier and enabling real-time monitoring and customizable thresholds.
- **Benefit**: Higher accuracy in temperature detection reduces false alarms and improves the reliability of fire hazard detection.

3. Microcontroller Integration

- **Need**: To offer advanced functionalities, such as customizable alarm settings, logging data, and interfacing with cloud services.
- **Enhancement**: Incorporating a microcontroller like an Arduino or ESP32 would enable features like data logging, programmable alarm thresholds, and integration with cloud services for historical alert tracking. The microcontroller also offers scalability for adding additional sensors (e.g., smoke detection).

- **Benefit**: The system becomes more intelligent and flexible, allowing for real-time data tracking, analysis, and easier customization.

4. Heat-Resistant, Fireproof Casing

- **Need**: Protection of internal components in extreme conditions.
- **Enhancement**: Enclosing the circuit in a heat-resistant and fireproof casing ensures the electronics are protected during a fire, maintaining the system's integrity and functionality.
- **Benefit**: Ensures long-term durability and operational safety, even in extreme fire situations.

CHAPTER 7

CONCLUSION

7.1 SUMMARY OF ACHIEVEMENTS

The fire alarm circuit successfully fulfils its primary objective of detecting high temperatures and triggering an audible alert to warn users of potential fire hazards. It demonstrates effective functionality using a thermistor-based sensing mechanism combined with a 555 Timer IC in astable mode to activate the buzzer. The system is simple, cost-effective, and reliable for basic fire detection applications. Its compact design and ease of implementation make it ideal for educational purposes and small-scale use. Overall, the project achieves its goal of creating a functional fire alarm system with clear scope for future enhancements, making it a solid foundation for more advanced safety solutions.

7.2 FINAL REMARK

In conclusion, this fire alarm circuit project has not only met its intended goal of detecting high temperatures and providing an audible warning but also laid the groundwork for future advancements in fire safety technology. Through the use of fundamental electronic components like the 555 Timer IC and thermistor, the project demonstrates a practical, cost-effective approach to fire detection. Moreover, the simplicity of the design ensures ease of replication and understanding, making it an excellent educational model. As highlighted in the future scope, this basic system holds vast potential for transformation into a smart, multi-functional safety device with modern features such as wireless alerts, digital sensors, and cloud integration. The project serves as a stepping stone toward more intelligent and responsive fire safety systems suitable for contemporary residential, commercial, and industrial environments.

ANNEXURES

DATASHEET

https://www.ti.com/lit/ds/symlink/ne555.pdf

TEST RESULTS

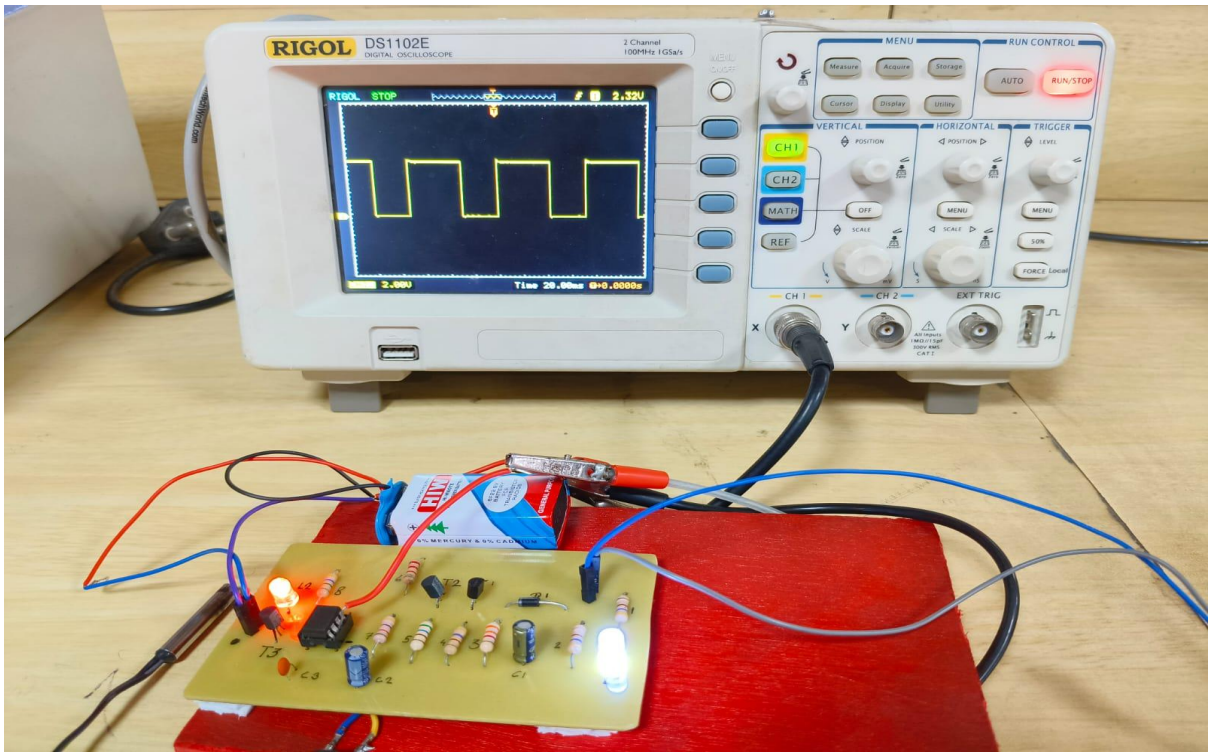


Fig 8.1 PCB Output Waveform

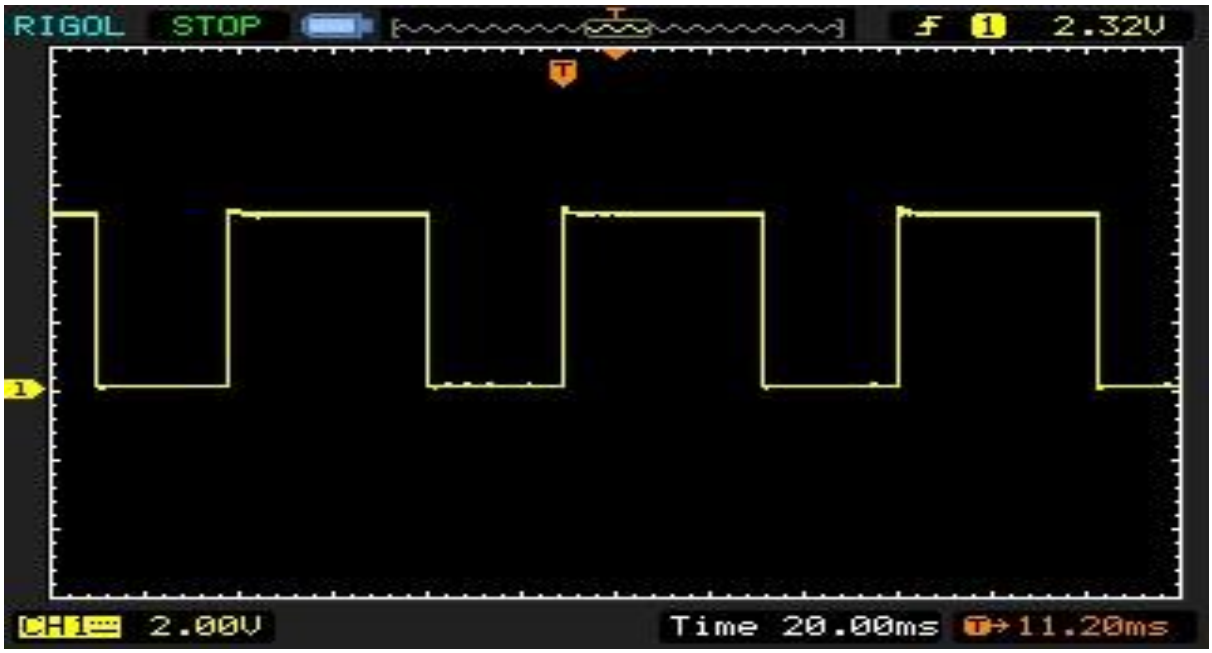


Fig 8.2 PCB Output

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